



Roll No. 111
NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD
DEPARTMENT OF COMPUTER SCIENCE
END-TERM EXAMINATION – FALL 2024 – BSCS VII (MORNING)

Subject: Compiler Construction
Allowed Time: 03 Hours

Instructor: Mr. Rana Abdul Sami Khan
Total Marks: 50

Instructions: -

- Attempt all questions and return the question paper with the answer sheet.
- Clearly mention the question number before attempting it

Part I

Consider the following CFG for Part I

$G \rightarrow S \mid a$
 $L \rightarrow (G) \mid p$
 $S \rightarrow bS \mid r$

QUESTION 1 [CLO-2, C3, PLO-3] 07 Marks

Carry out the concept of bottom up parsing and construct the Deterministic flow to construct shift reduced parsing.

QUESTION 2 [CLO-3, C2, PLO-3] 07+06 Marks

- Apply Shift reduced parsing to Construct SLR (1) Parsing table for the given rule
- Construct and perform stack processing for any of the complex sentence if possible using SLR (1) parsing table

QUESTION 3 [CLO-3, C5, PLO-3] 10 Marks

$M \rightarrow Ma \mid Mb \mid bR$
 $R \rightarrow (abR \mid Rab \mid ab)(C)$
 $C \rightarrow 0 \mid 1$

Construct LALR parsing for the above rule.

Part II

$F \rightarrow T \mid (P)$
 $P \rightarrow i \mid I, P$
 $T \rightarrow M \mid tM$
 $M \rightarrow bM \mid R$
 $R \rightarrow Rr \mid r$

QUESTION 4 [CLO-2, C3, PLO-3] 10 Marks

Apply top down parsing logics to write first() and follow() for all non-terminals and Apply top down parsing logics to write LL(1) Predictive parsing table for the given rule.

QUESTION 5 [CLO-3, C3, PLO-3] 10 Marks

Construct and perform stack processing activity for any of the possible supported sentence.

*Its not the only examination of your life...
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